

T3 UV Lagrangian terms grouped by field content

EFT order: 5; loop order: 1. Rows are model configurations and columns are field configurations. The BSM-only UV expression is preferred; the full UV Lagrangian is used as a fallback when the BSM-only conversion is unavailable.

Field combinations 1 of 4

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	$S_1 D$	S_2
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-S_1 S_1^\dagger M_{S_1}^2$ $+\frac{1}{2} S_1^2 \left(S_1^\dagger \right)^2 \lambda_{S_1}^{(1)}$	$+D_{\mu_1} (S_1)^\dagger D_{\mu_1} (S_1)$	$-M_{S_2}^2 S_2 S_2^\dagger$ $+\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}^1 \lambda_{S_2}^{(1)}$ $+\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}^2 \lambda_{S_2}^{(2)}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-S_{1,j_1} S_{1,j_1}^\dagger M_{S_1}^2$ $+\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}^1 \lambda_{S_1}^{(1)}$	$+D_{\mu_1} (S_{1,j_1})^\dagger D_{\mu_1} (S_{1,j_1})$	$-M_{S_2}^2 S_2 S_2^\dagger$ $+\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}^1 \lambda_{S_2}^{(1)}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-S_{1,j_1} S_{1,j_1}^\dagger M_{S_1}^2$ $+\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}^1 \lambda_{S_1}^{(1)}$	$+D_{\mu_1} (S_{1,j_1})^\dagger D_{\mu_1} (S_{1,j_1})$	$-M_{S_2}^2 S_2 S_2^\dagger$ $+\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}^1 \lambda_{S_2}^{(1)}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-S_{1,j_1} S_{1,j_1}^\dagger M_{S_1}^2$ $+\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}^1 \lambda_{S_1}^{(1)}$ $+\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}^2 \lambda_{S_1}^{(2)}$	$+D_{\mu_1} (S_{1,j_1})^\dagger D_{\mu_1} (S_{1,j_1})$	$-M_{S_2}^2 S_2 S_2^\dagger$ $+\frac{1}{2} S_2^2 \left(S_2^\dagger \right)^2 \lambda_{S_2}^{(1)}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-S_{1,j_1} S_{1,j_1}^\dagger M_{S_1}^2$ $+\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}^1 \lambda_{S_1}^{(1)}$ $+\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}^2 \lambda_{S_1}^{(2)}$	$+D_{\mu_1} (S_{1,j_1})^\dagger D_{\mu_1} (S_{1,j_1})$	$-M_{S_2}^2 S_2 S_2^\dagger$ $+\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}^1 \lambda_{S_2}^{(1)}$ $+\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}^2 \lambda_{S_2}^{(2)}$

Field combinations 2 of 4

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_2 D$	F	$F D$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+D_{\mu_1}(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$	$-\overline{F_{j_1,r_1}} F_{j_1,r_1} M_F$	$+i\overline{F_{j_1,r_1}} \gamma^{\mu_1} D_{\mu_1}(F_{j_1,r_1})$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+D_{\mu_1}(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$	$-\overline{F_{r_1}} F_{r_1} M_F$	$+i\overline{F_{r_1}} \gamma^{\mu_1} D_{\mu_1}(F_{r_1})$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+D_{\mu_1}(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$	$-\overline{F_{j_1,r_1}} F_{j_1,r_1} M_F$	$+i\overline{F_{j_1,r_1}} \gamma^{\mu_1} D_{\mu_1}(F_{j_1,r_1})$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+D_{\mu_1}(S_2)^\dagger D_{\mu_1}(S_2)$	$-\overline{F_{j_1,r_1}} F_{j_1,r_1} M_F$	$+i\overline{F_{j_1,r_1}} \gamma^{\mu_1} D_{\mu_1}(F_{j_1,r_1})$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+D_{\mu_1}(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$	$-\overline{F_{j_1,r_1}} F_{j_1,r_1} M_F$	$+i\overline{F_{j_1,r_1}} \gamma^{\mu_1} D_{\mu_1}(F_{j_1,r_1})$

Field combinations 3 of 4

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 H$	$S_1 S_2$	$S_2 H$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+H_{a_2} S_1 H_{a_1}^\dagger S_1^\dagger \mathcal{I}_{H1,a_1,a_2}^1 \lambda_{H1}^{(1)}$	$+S_1 S_{2,j_2} S_1^\dagger S_{2,j_1}^\dagger \mathcal{I}_{12,j_1,j_2}^1 \lambda_{12}^{(1)}$	$+\frac{1}{2} H_{a_2} S_{2,j_2} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}^1 \lambda_{H2}^{(1)}$ $+\frac{1}{2} H_{a_2} S_{2,j_2} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_2,j_1}^1 \lambda_{H2}^{(1)}$ $+\frac{1}{2} H_{a_2} S_{2,j_2} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}^2 \lambda_{H2}^{(2)}$ $+\frac{1}{2} H_{a_2} S_{2,j_2} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_2,j_1}^2 \lambda_{H2}^{(2)}$
cont.										
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+H_{a_2} S_{1,j_2} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}^1 \lambda_{H1}^{(1)}$ $+H_{a_2} S_{1,j_2} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}^2 \lambda_{H1}^{(2)}$	$+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^1 \lambda_{12}^{(1)}$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^2 \lambda_{12}^{(2)}$	$+H_{a_2} S_{2,j_2} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}^1 \lambda_{H2}^{(1)}$ $+H_{a_2} S_{2,j_2} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}^2 \lambda_{H2}^{(2)}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+H_{a_2} S_{1,j_2} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}^1 \lambda_{H1}^{(1)}$ $+H_{a_2} S_{1,j_2} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}^2 \lambda_{H1}^{(2)}$	$+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^1 \lambda_{12}^{(1)}$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^2 \lambda_{12}^{(2)}$	$+H_{a_2} S_{2,j_2} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}^1 \lambda_{H2}^{(1)}$ $+H_{a_2} S_{2,j_2} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}^2 \lambda_{H2}^{(2)}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+\frac{1}{2} H_{a_2} S_{1,j_2} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}^1 \lambda_{H1}^{(1)}$ $+\frac{1}{2} H_{a_2} S_{1,j_2} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}^2 \lambda_{H1}^{(2)}$ $+\frac{1}{2} H_{a_2} S_{1,j_2} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_2,j_1}^2 \lambda_{H1}^{(2)}$	$+S_2 S_{1,j_2} S_2^\dagger S_{1,j_1}^\dagger \mathcal{I}_{12,j_1,j_2}^1 \lambda_{12}^{(1)}$	$+H_{a_2} S_2 H_{a_1}^\dagger S_2^\dagger \mathcal{I}_{H2,a_1,a_2}^1 \lambda_{H2}^{(1)}$
cont.										
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{1}{2} H_{a_2} S_{1,j_2} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}^1 \lambda_{H1}^{(1)}$ $+\frac{1}{2} H_{a_2} S_{1,j_2} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_2,j_1}^1 \lambda_{H1}^{(1)}$ $+\frac{1}{2} H_{a_2} S_{1,j_2} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}^2 \lambda_{H1}^{(2)}$ $+\frac{1}{2} H_{a_2} S_{1,j_2} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H1,a_1,a_2,j_2,j_1}^2 \lambda_{H1}^{(2)}$	$+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^1 \lambda_{12}^{(1)}$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^2 \lambda_{12}^{(2)}$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^3 \lambda_{12}^{(3)}$	$+\frac{1}{2} H_{a_2} S_{2,j_2} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}^1 \lambda_{H2}^{(1)}$ $+\frac{1}{2} H_{a_2} S_{2,j_2} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_2,j_1}^1 \lambda_{H2}^{(1)}$ $+\frac{1}{2} H_{a_2} S_{2,j_2} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}^2 \lambda_{H2}^{(2)}$ $+\frac{1}{2} H_{a_2} S_{2,j_2} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H2,a_1,a_2,j_2,j_1}^2 \lambda_{H2}^{(2)}$

Field combinations 4 of 4

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 S_2 H$	$S_1 F \ell$	$S_2 F \ell$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+\overline{\lambda_{T3} S_{2,j1}} \mathcal{I}_{T3 a_1, a_2, j1}^* H_{a_1}^\dagger H_{a_2}^\dagger S_1^\dagger$	$+\overline{L_{a_1, i_1}} C P_R \overline{F_{j1, r1}}^T S_1 y_{1, i_1, r1} \mathcal{I}_{y1, a_1, j1}$	$+\overline{y_{2, i_1, r1}} \overline{F_{j1, r1}} P_L L_{a_1, i_1} S_{2, j1} \mathcal{I}_{y2 a_1, j1}^*$
								$+H_{a_1} H_{a_2} S_1 \mathcal{I}_{T3, a_1, a_2, j1} \lambda_{T3} S_{2, j1}^\dagger$	$+\overline{y_{1, i_1, r1}} L_{a_1, i_1}^T C P_L F_{j1, r1} \mathcal{I}_{y1 a_1, j1}^* S_1^\dagger$	$+\overline{L_{a_1, i_1}} P_R F_{j1, r1} y_{2, i_1, r1} \mathcal{I}_{y2, a_1, j1, j1} S_{2, j1}^\dagger$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+\overline{\lambda_{T3} S_{2, j2}} \mathcal{I}_{T3 a_1, a_2, j1, j2}^* H_{a_1}^\dagger H_{a_2}^\dagger S_{1, j1}^\dagger$	$+\overline{L_{a_1, i_1}} C P_R \overline{F_{r1}}^T S_{1, j1} y_{1, i_1, r1} \mathcal{I}_{y1, a_1, j1}$	$+\overline{y_{2, i_1, r1}} \overline{F_{r1}} P_L L_{a_1, i_1} S_{2, j1} \mathcal{I}_{y2 a_1, j1}^*$
								$+H_{a_1} H_{a_2} S_{1, j2} \mathcal{I}_{T3, a_1, a_2, j2, j1} \lambda_{T3} S_{2, j1}^\dagger$	$+\overline{y_{1, i_1, r1}} L_{a_1, i_1}^T C P_L F_{r1} \mathcal{I}_{y1 a_1, j1}^* S_{1, j1}^\dagger$	$+\overline{L_{a_1, i_1}} P_R F_{r1} y_{2, i_1, r1} \mathcal{I}_{y2, a_1, j1} S_{2, j1}^\dagger$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+\overline{\lambda_{T3} S_{2, j2}} \mathcal{I}_{T3 a_1, a_2, j1, j2}^* H_{a_1}^\dagger H_{a_2}^\dagger S_{1, j1}^\dagger$	$+\overline{L_{a_1, i_1}} C P_R \overline{F_{j1, r1}}^T S_{1, j1} y_{1, i_1, r1} \mathcal{I}_{y1, a_1, j1, j1}$	$+\overline{y_{2, i_1, r1}} \overline{F_{j1, r1}} P_L L_{a_1, i_1} S_{2, j1} \mathcal{I}_{y2 a_1, j1, j1}^*$
								$+H_{a_1} H_{a_2} S_{1, j2} \mathcal{I}_{T3, a_1, a_2, j2, j1} \lambda_{T3} S_{2, j1}^\dagger$	$+\overline{y_{1, i_1, r1}} L_{a_1, i_1}^T C P_L F_{j1, r1} \mathcal{I}_{y1 a_1, j1, j1}^* S_{1, j1}^\dagger$	$+\overline{L_{a_1, i_1}} P_R F_{j1, r1} y_{2, i_1, r1} \mathcal{I}_{y2, a_1, j1, j1} S_{2, j1}^\dagger$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+H_{a_1} H_{a_2} S_{1, j1} \mathcal{I}_{T3, a_1, a_2, j1} \lambda_{T3} S_2^\dagger$	$+\overline{L_{a_1, i_1}} C P_R \overline{F_{j1, r1}}^T S_{1, j1} y_{1, i_1, r1} \mathcal{I}_{y1, a_1, j1, j1}$	$+\overline{y_{2, i_1, r1}} \overline{F_{j1, r1}} P_L L_{a_1, i_1} S_2 \mathcal{I}_{y2 a_1, j1}^*$
								$+\overline{\lambda_{T3} S_2} \mathcal{I}_{T3 a_1, a_2, j1}^* H_{a_1}^\dagger H_{a_2}^\dagger S_{1, j1}^\dagger$	$+\overline{y_{1, i_1, r1}} L_{a_1, i_1}^T C P_L F_{j1, r1} \mathcal{I}_{y1 a_1, j1, j1}^* S_{1, j1}^\dagger$	$+\overline{L_{a_1, i_1}} P_R F_{j1, r1} y_{2, i_1, r1} \mathcal{I}_{y2, a_1, j1} S_2^\dagger$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\overline{\lambda_{T3} S_{2, j2}} \mathcal{I}_{T3 a_1, a_2, j1, j2}^* H_{a_1}^\dagger H_{a_2}^\dagger S_{1, j1}^\dagger$	$+\overline{L_{a_1, i_1}} C P_R \overline{F_{j1, r1}}^T S_{1, j1} y_{1, i_1, r1} \mathcal{I}_{y1, a_1, j1, j1}$	$+\overline{y_{2, i_1, r1}} \overline{F_{j1, r1}} P_L L_{a_1, i_1} S_{2, j1} \mathcal{I}_{y2 a_1, j1, j1}^*$
								$+H_{a_1} H_{a_2} S_{1, j2} \mathcal{I}_{T3, a_1, a_2, j2, j1} \lambda_{T3} S_{2, j1}^\dagger$	$+\overline{y_{1, i_1, r1}} L_{a_1, i_1}^T C P_L F_{j1, r1} \mathcal{I}_{y1 a_1, j1, j1}^* S_{1, j1}^\dagger$	$+\overline{L_{a_1, i_1}} P_R F_{j1, r1} y_{2, i_1, r1} \mathcal{I}_{y2, a_1, j1, j1} S_{2, j1}^\dagger$